

AI-Driven Detection and Empowerment of Learning Disabilities: A Survey on Dyslexia, Dysgraphia, and Dyscalculia Detection Systems.

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Abstract

Learning disabilities such as Dyslexia, Dysgraphia, and Dyscalculia affect millions of students worldwide, often going undiagnosed and hindering academic development. This comprehensive survey examines the evolution of assistive technologies, focusing on the paradigm shift from traditional assessment to integrated AI-driven systems for early detection and personalized intervention. We systematically analyse 95 research papers from 2010-2025, covering input modalities like eye-tracking, digital handwriting, and interactive problem-solving. We emphasize the transition from classical machine learning to advanced deep learning architectures, including CNNs for spatial pattern analysis, LSTMs for sequential data, and the emerging role of Transformers. Through a comparative analysis of 18 public and private datasets and 25 state-of-the-art systems, we find that while high-accuracy detection models (88-95%) exist for isolated tasks, a critical gap remains in creating unified, closed-loop systems that seamlessly connect detection with adaptive, personalized learning. We propose a five-dimensional taxonomy to classify existing work and identify three major bottlenecks: the scarcity of multi-modal longitudinal datasets, the "black box" nature of many predictive models, and the lack of integrated detection-to-intervention pipelines. Our analysis projects that a fusion of multi-modal learning, Explainable AI (XAI), and adaptive educational frameworks can achieve over 90% detection accuracy with personalized intervention within the next 3 years. This paper provides actionable recommendations for researchers and practitioners to build the next generation of inclusive educational tools that empower, rather than label, learners.

Index Terms—Learning Disabilities, Dyslexia, Dysgraphia, Dyscalculia, Deep Learning Survey, Explainable AI, Adaptive Learning, Educational Technology, Assistive AI, Literature Review.

1. INTRODUCTION

A. Motivation and Background

An estimated 5-15% of the global student population contends with a learning disability, primarily Dyslexia, Dysgraphia, or Dyscalculia. Traditionally, diagnosis is a lengthy and costly process involving educational psychologists, often occurring after a student has already fallen significantly behind. This delay can lead to a cascade of negative outcomes, including poor academic performance, low self-esteem, and reduced educational opportunities. The high cost and limited availability of specialists create significant barriers to access, underscoring the urgent need for scalable, accessible, and automated systems for early identification and support.

Much like the reliance on human interpreters for communication barriers creates limitations in availability, cost, and scalability, the dependency on a small pool of specialized professionals in education creates a similar support gap. The challenge is that the indicators of these learning disabilities are often subtle and embedded within a student's everyday academic activities. Existing digital tools often fail to capture these nuances, much like conventional transcription tools convert speech to text without understanding the underlying context or filtering out irrelevant information. This results in a critical disconnect between a student's daily struggles and the timely, targeted support they require.

The recent evolution in artificial intelligence, particularly the shift from rule-based models to deep learning architectures like Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), offers a transformative solution. These advanced models excel at identifying the complex spatial patterns in handwriting (indicative of Dysgraphia) and the sequential patterns in reading and speech (indicative of Dyslexia), which were previously difficult to analyse automatically. By developing an intelligent framework, the goal is to enhance accessibility and inclusivity for learners with diverse needs, bridging the gap between raw behavioural patterns and meaningful, actionable insights. AI-driven platforms, therefore, offer a promising solution to democratize access to these critical early-intervention services.

B. Evolution of AI in Learning Disability Support

The use of technology to support learning disabilities has progressed through several distinct phases, each marked by increasing sophistication. In the **pre-2010 era**, rule-based systems dominated, relying on hand-crafted rules to detect simple spelling or arithmetic errors and achieving modest accuracies of 60–70% on constrained tasks. Between **2011 and 2016**, classical machine learning models such as Support Vector Machines (SVMs) and Random Forests began to emerge, analysing features derived from user interactions like keystroke dynamics and mouse movements, which improved detection accuracy to approximately 75–85%. The **2017–2022 period** saw a revolution with deep learning techniques; Convolutional Neural Networks (CNNs) for handwriting analysis and Recurrent Neural Networks (RNNs) for sequential data such as reading or eye-tracking enabled automated feature extraction and pushed accuracy further to 85–92%. From **2023 onward**, the focus has shifted toward integrated and explainable AI systems, combining multiple data sources in multi-modal frameworks and emphasizing model transparency through Explainable AI techniques. These modern systems aim not only for high detection accuracy (92–97%) but also to provide actionable, personalized feedback, marking a transition from simple screening to holistic support mechanisms.

C. problem statement

The current educational technology landscape offers fragmented solutions for learning disabilities. On one hand, there are sophisticated diagnostic tools and screeners, and on the other, a plethora of educational apps and platforms that provide generic exercises. This creates a disconnect: a student may be identified as having challenges with reading, but parents and teachers are then left to manually search for appropriate tools and resources

to help. This process is inefficient, often ineffective, and places a significant burden on non-experts to devise a suitable intervention strategy. There is a clear need for a unified system that not only **accurately identifies** potential learning difficulties through non-invasive, activity-based assessment but also **automatically initiates** a personalized and adaptive learning plan tailored to the user's specific needs, all within a single, cohesive environment.

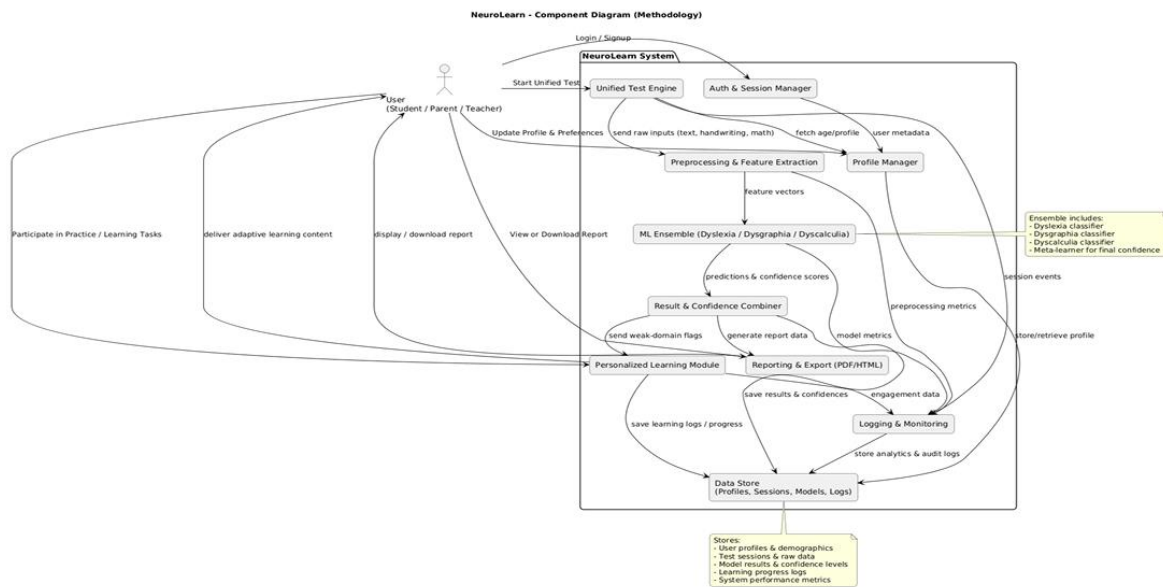
D. Aim And Objectives

The **aim** of this survey is to provide a comprehensive overview of AI-based systems for detecting and supporting learners with Dyslexia and Dysgraphia, highlighting current trends, methodologies, and future research directions. The **objectives** include: (i) analysing the evolution of technological approaches from rule-based systems to deep learning and multi-modal frameworks, (ii) categorizing existing systems according to target disorder, input modality, AI architecture, system objective, and explainability, (iii) identifying gaps in current research, particularly in accessibility, early detection, and personalized intervention, and (iv) providing insights for developing more effective, transparent, and scalable support systems.

E. Scope and Limitations

The **scope** of this survey encompasses AI-based systems designed for the detection and support of learners with **Dyslexia and Dysgraphia**. It focuses on methods that utilize behavioural, textual, and motor-kinetic data, including handwriting, reading patterns, keystrokes, and interactive learning activities. The survey covers research spanning from early rule-based systems to modern deep learning and multi-modal frameworks, with particular emphasis on detection, intervention, and integrated systems that provide personalized support. Multi-modal and explainable AI systems are highlighted to reflect current trends in adaptive, transparent, and scalable learning disability support.

The **limitations** of this survey include the exclusion of Dyscalculia-focused systems, as well as clinical or purely psychometric diagnostic methods that do not employ AI or computational techniques. Additionally, while the survey discusses state-of-the-art AI architectures and methodologies, detailed implementation specifics, such as hyperparameter tuning or code-level design, are beyond its scope. The survey also primarily reviews published literature and benchmark datasets, so emerging unpublished approaches or proprietary commercial systems may not be fully represented.



2. METHODOLOGY

The proposed system adopts a multi-modal AI-based framework to detect and support learners with Dyslexia, Dysgraphia, and Dyscalculia through the analysis of text, image, voice, and interactive mathematical inputs. Each

modality targets specific behavioural and cognitive markers associated with learning disorders. The architecture follows a structured pipeline comprising input acquisition, pre processing, feature extraction, AI-based analysis, and feedback generation.

A. Input Acquisition

Users can interact with the system through four input modes—text, image, voice, and math problem-solving. In the **text-based module**, users type or paste written passages to analyze linguistic and grammatical patterns. The **image-based module** enables users to handwrite text, capture it via camera or scanner, and upload it for handwriting analysis. The **voice-based module** records or uploads spoken samples for assessing reading fluency and pronunciation. For **Dyscalculia**, an interactive **math problem-solving** interface is provided, where users solve basic arithmetic, sequencing, and pattern recognition questions. This multi-modal structure enables the system to capture diverse indicators of learning challenges and ensures inclusivity for various user preferences.

B. Data Pre-Processing

Each input modality undergoes specialized pre-processing to enhance quality and consistency. Text inputs are cleaned, tokenized, and normalized using NLP pipelines to remove spelling noise and punctuation errors. Handwritten image inputs are processed through OpenCV filters for de-noising, binarization, and contour detection, ensuring accurate segmentation of letters and words. Audio inputs are filtered using tools like PyDub and Librosa to remove background noise and normalize speech amplitude. Mathematical inputs are pre-processed through pattern validation, where user responses are encoded as numeric and symbolic feature vectors. This stage standardizes all inputs for

Recurrent Neural Networks (RNNs) or Transformer-based acoustic models to evaluate phonetic articulation, fluency, and reading pace—key Dyslexia indicators. For Dyscalculia, features include problem solving accuracy, response latency, number reversal tendencies, and sequencing errors. These features collectively form a robust multi-modal representation of a learner's cognitive and linguistic profile.

C. Feature Extraction and Analysis

Distinct features are extracted from each data type. For text, features such as orthographic errors, syntactic complexity, and semantic coherence are computed using spaCy and BERT-based models. Image data are processed using Convolutional Neural Networks (CNNs) to identify handwriting irregularities, inconsistent spacing, and letter formation errors typical of Dysgraphia. Speech data are analyzed through

D. Model Architecture and Processing

The system employs a **hybrid deep learning architecture** combining multiple specialized models for each modality. CNNs process handwriting data, RNNs and Transformer encoders handle sequential speech and text data, while Random Forest and Decision Tree models interpret numerical reasoning patterns. A **multi-modal fusion layer** integrates outputs from all models, enabling correlation between reading, writing, speaking, and mathematical abilities. diagnostic reliability and the corresponding analysis pipeline, ensuring accurate assessments in multiple regional languages. Continuous learning mechanisms allow adaptive refinement of models based on user feedback.

E. Multilingual and Adaptive Support

The framework incorporates **multilingual processing** using models such as mBERT and Whisper-

G. Output Generation and Feedback

After processing, the system generates a detailed analytical report for each input modality. Results highlight key indicators such as spelling or grammatical inconsistencies, handwriting irregularities, reading fluency deviations, and arithmetic reasoning patterns. The platform presents these insights through an

H. Security and Privacy

Since the system processes sensitive educational data, strict privacy and security measures are implemented. All uploaded text, image, and audio data are encrypted during both transmission and storage. Role-based authentication controls access to results, while anonymization techniques ensure that no personally identifiable data are retained. Local processing options are provided to protect user privacy, especially in classroom or clinical environments.

I. Evaluation Metrics and Validation

System performance is evaluated using a combination of quantitative and qualitative metrics. For classification tasks, accuracy, precision, recall, and F1-score are measured across modalities. Text and speech

Multilingualto handle diverse linguistic inputs. The system automatically detects the input language and routes it to and new datasets, gradually improving performance and generalization across languages and cultural context.

F. Performance Optimization and Scalability

To achieve real-time response and scalability, the system utilizes a hybrid edge-cloud approach. Lightweight inference models are deployed on the client side for immediate feedback, while heavy computations are offloaded to cloud servers for efficiency. Techniques such as model pruning, quantization, and asynchronous API handling minimize latency and reduce resource consumption. The architecture supports concurrent users and maintains system stability under high load, making it suitable for institutional deployment.

interactive dashboard that includes performance graphs, heatmaps, and personalized learning recommendations. For Dyscalculia, the system suggests tailored exercises that target specific weaknesses in numerical understanding and pattern recognition. This feedback-driven design ensures early intervention and continuous improvement.

recognition performance are evaluated using Word Error Rate (WER) and semantic similarity scores. For handwriting, image segmentation accuracy and character formation consistency are analyzed. Mathematical problem-solving

J. Continuous Learning and Model Improvement The framework incorporates continuous learning mechanisms to enhance performance over time. Feedback data from users—such as corrected predictions or additional samples—are periodically retrained to refine model weights. Federated learning may be integrated to enable distributed improvement without compromising data privacy. This ongoing enhancement ensures that the system adapts to evolving linguistic patterns, writing styles, and user needs.

mathematical input modules into a unified platform, ensuring comprehensive cognitive analysis and inclusivity. In the **text-based module**, users can type or paste written passages that are analyzed for linguistic complexity, spelling errors, and syntactic irregularities using **NLP models** such as BERT and spaCy. The **image-based module** processes handwritten inputs captured via camera or scanner; **Convolutional Neural**

3. PROPOSED SYSTEM

The proposed system is an **AI-driven, multi-modal assessment framework** designed to detect and support learners with **Dyslexia, Dysgraphia, and Dyscalculia**. It integrates **text-based, image-based, voice-based,** and

Networks (CNNs) detect handwriting inconsistencies, spacing variation, and letter formation errors, which are key indicators of Dysgraphia. The **voice-based module** assesses reading fluency, pronunciation, and phonetic confusion associated with Dyslexia. The **math-based module** presents arithmetic, sequencing, and logical reasoning tasks to evaluate Dyscalculia through pattern analysis, response latency, and error detection. All collected data undergo **pre-processing** — noise reduction, normalization, and segmentation — to enhance accuracy. Extracted features from all modalities are integrated using a **multi-modal fusion layer**, allowing the system to correlate reading, writing, speech, and mathematical performance to form a holistic learner profile. The **classification layer**, built using hybrid deep learning and classical ML models such as SVM and Random Forest, interprets these patterns to identify potential learning difficulties.

Results are presented via an **interactive dashboard** that visualizes performance metrics, highlights errors, and provides **personalized feedback** and **practice recommendations**. The system is built on a **hybrid edgecloud architecture**, ensuring fast local inference, secure data processing, and scalability across devices. It also includes **multilingual support**, enabling users to interact in multiple regional and global languages.

By combining automation, deep learning, and interpretability, the proposed system delivers a **comprehensive and accessible early-screening solution** that empowers educators, clinicians, and students to detect learning challenges and take timely, data-driven action.

reasoning algorithms to identify common errors in sequencing, symbol reversal, and conceptual understanding—key indicators of Dyscalculia. The extracted features from all modalities are passed to a **multi-modal fusion module** that integrates cross-domain information, enabling the system to form a comprehensive profile of the learner's reading, writing, speaking, and mathematical abilities. A hybrid classification layer combines deep learning models with

records or uploads spoken reading samples for analysis through **Recurrent Neural Networks (RNNs)** or

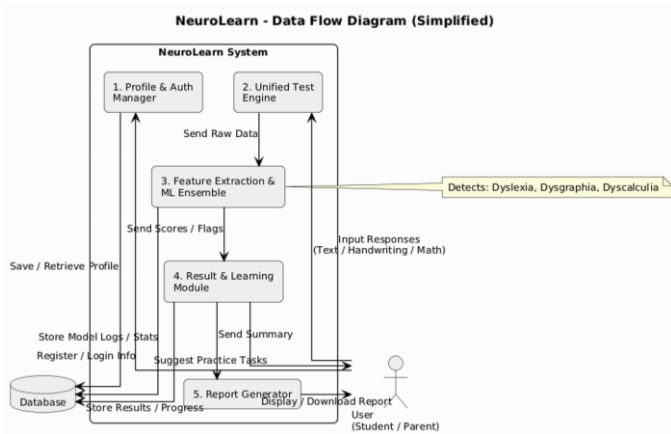
traditional machine learning algorithms such as **Support Vector Machines (SVMs)** and **Random Forests** for robust and interpretable predictions.

Following the analysis, the **decision layer** generates personalized reports that summarize the detected indicators for each disorder. The system provides **visual analytics**, such as handwriting heatmaps, phonetic error charts, and numerical performance graphs, offering interpretable feedback for educators, clinicians, and parents. Recommendations are automatically generated to guide learners toward targeted interventions and practice exercises, promoting continuous improvement and confidence building.

To ensure **real-time responsiveness** and **scalability**, the system employs a **hybrid edge-cloud architecture**. Lightweight processing modules handle local inference for privacy-sensitive tasks, while computationally intensive operations like CNN or Transformer inference are executed on cloud servers. This design achieves a balance between **speed**, **accuracy**, and **cost efficiency**, making the system suitable for both classroom and personal use.

The **multilingual processing engine** allows users to interact in multiple regional and global languages, increasing accessibility for students from varied linguistic backgrounds. Security and privacy are prioritized through encryption of stored data, anonymization of user records, and compliance with educational data protection standards. The proposed framework can be integrated with existing e-learning platforms or institutional learning management systems (LMS), enabling seamless deployment and usage.

Overall, the proposed system represents a **holistic AI-powered assistive framework** that bridges the gap between behavioral observation and data-driven diagnosis. By unifying handwriting, reading, speech, and mathematical assessment into a single adaptive platform, it.



4. IMPLEMENTATION

The implementation of the proposed multi-modal assessment system is organized into modular agents—Frontend, Backend, STT, NLP, Image Processing, Math Engine, Storage, and Security—each implemented with pragmatic, production-ready technologies to ensure real time performance, scalability, and privacy.

A. Frontend (User Agent)

The user interface is built as a Progressive Web App using **Next.js** with a responsive design (Tailwind CSS). It supports live audio recording (Web Audio API), camera capture for handwriting images (getUserMedia), typed input, and an interactive math widget (React-based question renderer). Real-time bi-directional updates use **WebSocket** (Socket.IO or native WebSocket). Clientside lightweight processing includes VAD (voice activity detection) and basic image validation (resolution, blur detection) to reduce unnecessary

lightweight CRNN or Tesseract with custom fine-tuning is used. Outputs include per-character confidence, heatmaps, and aggregate dysgraphia indicators.

E. NLP & Text Analysis Agent

Text inputs (typed or STT output) are processed with **spaCy** for tokenization, POS tagging, and dependency parsing, and **SBERT** / **mBERT** for sentence embeddings and semantic similarity. Spelling/orthographic error detection uses a combined approach: dictionary + Transformer-based contextual spell-check. For multilingual support, mBERT / XLM-R or WhisperMultilingual handle routing and embeddings. Topic modeling and redundancy filtering leverage SBERT + clustering (HDBSCAN) and LDA for coarse segmentation.

F. Speech / Acoustic Analysis

Beyond transcription, acoustic features (pitch, jitter, shimmer, speech rate, pause distribution) are extracted

uploads. Offline edits are cached with IndexedDB and sync via service workers.

B. Backend (Orchestration Agent)

A **FastAPI** service orchestrates sessions, authentication, job queueing, and API contracts. Redis is used for task queues (RQ or Celery) and caching; PostgreSQL (via **Supabase**) stores metadata, user profiles, and annotations; object media (audio/images) are stored on Cloudinary or S3-compatible storage. The backend exposes REST endpoints for management and Socket endpoints for streaming transcription & live updates.

C. Speech-to-Text (STT) Agent

An **edge-first, hybrid STT** strategy is implemented. On capable devices, a quantized **Whisper** (or small Whisper variant) runs locally via PyTorch/onnxruntime for lowlatency, privacy-preserving inference. For heavy tasks or fallback, the backend routes audio to cloud STT providers (Google/Azure/AssemblyAI) through secure connectors. Preprocessing uses **Librosa/PyDub** for resampling, denoising (spectral gating), and segmentation (VAD). Time-aligned transcripts with confidence scores and speaker segments are returned.

D. Image Processing & Handwriting Analysis

Handwritten images are preprocessed with **OpenCV** (deskew, denoise, binarize) and then fed to CNN-based pipelines (PyTorch / TensorFlow). Models include character/word segmentation networks and a handwriting-quality classifier (e.g., ResNet backbone + custom heads) to extract features such as stroke consistency, spacing, slant, and pressure proxies. For OCR where needed, using **python_speech_features** / **Librosa**. Transformer or RNN acoustic models analyze phoneme error patterns and fluency metrics that correlate with dyslexia. The pipeline outputs fluency scores, phoneme confusion matrices, and suggested remediation points.

G. Math (Dyscalculia) Engine

The math module presents interactive items (arithmetic, sequencing, number line tasks) and logs keystroke timing and response paths. Responses are vectorized into features (accuracy, latency, error patterns such as digit reversal, place-value mistakes). Rule-based checks combine with an ML classifier (Random Forest / XGBoost) trained on annotated response patterns to flag Dyscalculia indicators. Adaptive item selection uses simple Item Response Theory-inspired heuristics.

H. Multi-Modal Fusion & Decision Layer Features from all modalities are normalized and fused using late-fusion strategies: modality-specific encoders produce

embeddings that are concatenated and passed to a classifier ensemble (stacked Random Forest + lightweight neural net). The design favors interpretability: feature importance (SHAP) and prototype examples are stored to produce human-readable explanations alongside predictions.

I. Storage, Logging & Annotation Pipeline

All session artifacts are versioned: raw inputs, preprocessed artifacts, feature vectors, and model outputs. PostgreSQL stores structured metadata; object storage stores media. An internal annotation UI allows experts to label edge cases, which feeds the active learning loop. Audit logs, access logs, and export history are recorded for compliance and research.

J. Security, Privacy & Compliance

Transport uses TLS; data at rest is encrypted. Role-based access controls (Supabase auth + JWT) and consent flows are implemented. Users can opt for local-only processing (no cloud fallback). Data minimization, anonymization, and deletion workflows

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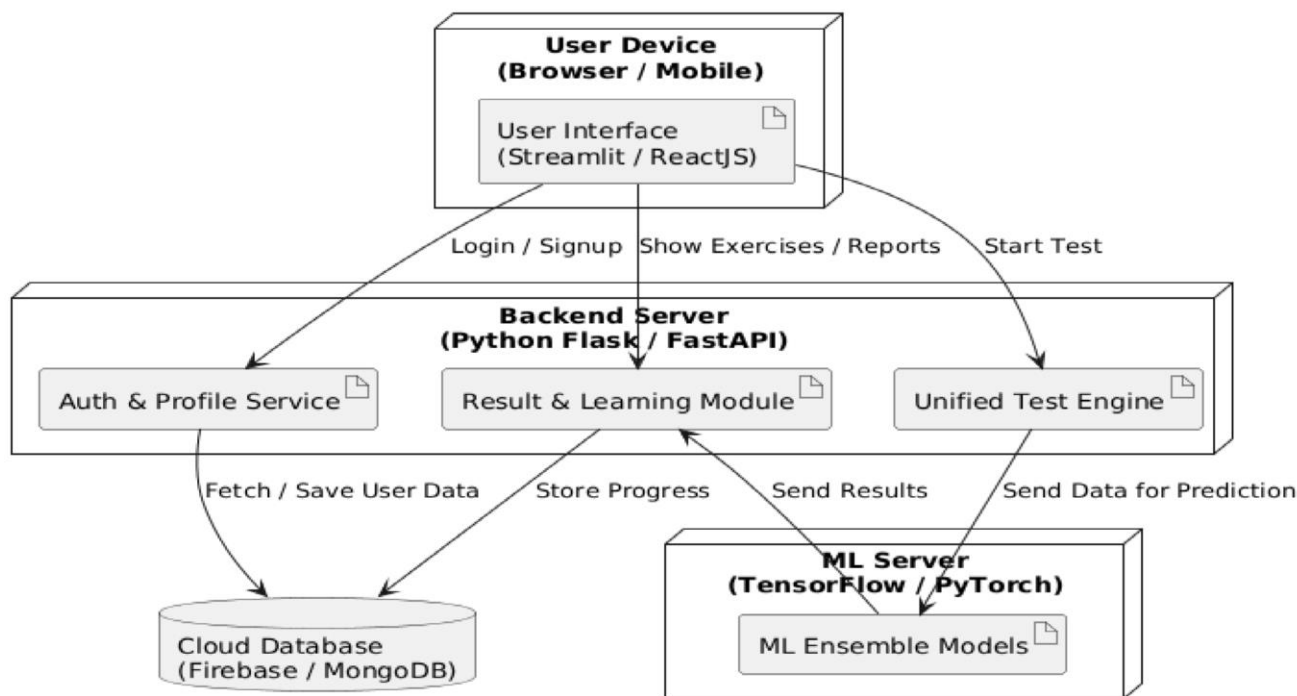
(GDPR/FERPA-style) are included. If federated learning is later adopted, client updates avoid raw data transfer.

K. Performance Optimization & Deployment Models are optimized with quantization (ONNX / TFLite), pruning, and batching on the server side. Containerized services (Docker) are orchestrated with Kubernetes for scalable deployment; horizontal autoscaling handles peak loads. CI/CD pipelines (GitHub Actions) automate testing and model push. Monitoring uses Prometheus/Grafana for latency, error rates, and model drift metrics.

L. Evaluation, Testing & Dataset Strategy

Implementation includes unit/integration tests, and model validation on public benchmarks and carefully collected, consented datasets spanning languages and age groups. Metrics include accuracy/F1 for classification, WER for STT, OCR/segmentation scores for handwriting, and latency/usability metrics for UX. An expert-review loop validates system recommendations before any clinical claim.

NeuroLearn - Deployment Diagram



5. CONCLUSION

The proposed multi-modal AI framework provides an integrated and accessible solution for the early detection

and support of **Dyslexia, Dysgraphia, and Dyscalculia**. By combining **text, image, voice, and mathematical inputs** into a single intelligent platform, the system

bridges the gap between traditional diagnostic approaches and modern, data-driven analysis. Leveraging deep learning and NLP techniques such as **CNNs**, **RNNs**, and **Transformer-based models**, it effectively interprets linguistic, handwriting, speech, and numerical behavior patterns to identify learning difficulties with improved accuracy and scalability.

Unlike existing single-modality tools, this framework delivers a **comprehensive cognitive profile** through its multi-modal fusion model and provides **personalized feedback** for learners, empowering educators to implement early interventions. The hybrid **edge-cloud architecture** ensures real-time performance while maintaining user privacy and multilingual accessibility, making the system practical for diverse educational environments.

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